

Naufal Altaf

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Summary

Information Systems graduate building software since 2019, primarily focused on full-stack web development. Experienced with Android platform development, machine learning, and cloud technologies through research and independent projects. Enjoys understanding software beyond the application layer and solving engineering problems across different domains.

Education

Mercu Buana University

Jakarta, Indonesia

Bachelor of Science, Information Systems.

May 2026

GPA 3.66. | Graduated in 3.5 years.

Relevant Coursework: Database Design, Java Programming, Python, Data Structures, Machine Learning.

Professional Experience

Developer and Researcher

March 2025

IEEE International Conference (ICOT 2025)

- Published Heart Disease Prediction Model using SEL-NNML Framework at an IEEE international conference.
- Built a stacking ensemble learning model combining deep learning with traditional machine learning classifiers including Random Forest and XGBoost.
- Developed data preprocessing and hyperparameter optimization pipelines to improve model generalization.
- Collaborated in a three-member research team from experimentation through publication.

Mobile Development Cohort

Aug 2024 – Dec 2024

Bangkit Academy (by Google, GoTo, Tokopedia & Traveloka)

- Developed Plantanist, an Android application for plant disease detection using on-device machine learning.
- Built the application in Kotlin following MVVM architecture and SOLID principles.
- Integrated TensorFlow models for real-time image classification.
- Collaborated with Mobile, ML and Cloud Computing cohorts.

Developer / Designer

Jun 2023 – Nov 2023

Program Kreativitas Mahasiswa (SIMBELMAWA)

- Designed a smart glasses HUD concept for real-time navigation.
- Produced system concepts and user flows.
- Collaborated with multi-disciplinary team members.

Selected Engineering Projects

Android Open-Source Project

- Developed a Xiaomi 15 (dada) custom ROM device tree by adapting open-source Android platform implementations.
- Resolved various issues leveraging information from debugging tools such as logcat, dmesg and dumpsys.
- Built AOSP and the Linux kernel to understand Android boot process, init, HALs and os-kernel interactions.
- Managed development using Git with structured commit history and collaborative workflows.

Full-Stack Web Applications

- Built and maintained web applications using Next.js, Python/Flask and RESTful APIs.
- Designed reusable frontend components and backend services.
- Applied practices such as SSR and CSR rendering where appropriate.
- Deployed applications using Vercel, Firebase and Google Cloud Platform.

Skills & Interests

- **Technical:** Python, TypeScript, Kotlin, Java, C++, React, Next.js, Tailwind CSS, Python, Flask, RESTful APIs, Google Cloud Platform, Firebase, Docker, Vercel, PyTorch, TensorFlow Scikit-Learn, NumPy, Linux, Bash, Git, Gerrit, AOSP, CUDA.
- **Certifications:** Cisco CCNA v7, Oracle Java Programming, Oracle Database Design, Dicoding Intermediate Android.
- **Language:** Native Indonesian and Fluent English.